

Coding & Solution

SMARTBRIDGE - Coding & Solution

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-6970
- **Team Size:** 3
- **Team Leader:** Prince Kumar P
- **Team Members:** Gowtham L, Guru Moorthi M
- **Project Name:** AI FitTrack
- **Maximum Marks:** 5 Marks

Solution Summary

Field	Details
Repository Link / URL	[https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y] (https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y)
Programming Language(s)	JavaScript (Node.js runtime environment v16+)
Framework(s) Used	Express.js (REST API framework), Mongoose ODM
Key Features Implemented	• JWT User Registration, Login & Profile Retrieval. • Full Workout CRUD Management (Create, Read, Search, Update, Delete).

	• Google Gemini AI Workout Recommendations & Performance Insights. • Centralized JSON Error Handling & Schema Validations.
Pending / Incomplete Features	None. All planned backend REST endpoints and AI features are 100% operational.
Setup / Run Instructions	1. Clone repository and navigate into project directory: <code>cd Code Files</code>. 2. Install dependencies: <code>npm install</code>. 3. Configure <code>.env</code> file with <code>PORT</code>, <code>MONGO_URI</code>, <code>JWT_SECRET</code>, and <code>GEMINI_API_KEY</code>. 4. Start server: <code>npm start</code>. 5. Execute API tests via Thunder Client / Postman at <code>http://localhost:5000</code>.

Code Quality Checklist

S.No	Criteria PDF	Status (Yes / No) PDF

1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments: Application verified successfully using Thunder Client. API endpoints handle database operations and Gemini AI requests within target response times.